Introducing Polynomials and the Factor and Remainder Theorems — Problem Set

Warm-up

1. Given $P(x) = x^3 - 2x^2 + 5x + 1$, find the value of $P(1)$.
2. Determine the remainder when $P(x) = x^2 + 4x - 7$ is divided by $(x - 3)$ using the remainder theorem.
3. If $P(2) = 0$, which of the following is a factor of $P(x)$?
   (A) $x - 1$
   (B) x
   (C) $x + 2$
   (D) $x - 2$
4. State the degree of the polynomial $P(x) = (x^2 + 1)(x - 5)^3$.
5. Find the remainder when $P(x) = 2x^3 - x + 4$ is divided by $(x + 1)$.
6. Show that $(x - 4)$ is a factor of $P(x) = x^2 - 2x - 8$.
7. Given $P(x) = x^3 + kx - 10$, find $P(0)$.
8. If $(x + 5)$ is a factor of $Q(x)$, what is the value of $Q(-5)$?
9. Simplify the expression for $h(x) = f(x) + g(x)$ where $f(x) = x^2 - 3$ and $g(x) = 2x + 1$.
10. Find the remainder when $x^4 + 5$ is divided by $(x - 1)$.

Skill-building

1. The polynomial $P(x) = x^3 + kx^2 - 4$ has a remainder of 8 when divided by $(x - 2)$. Find the value of k.
2. Show that $(x + 2)$ is a factor of $P(x) = x^3 + 3x^2 - 4$, and hence express $P(x)$ as a product of a linear and a quadratic factor.
3. Find the values of m and n such that $P(x) = x^3 + mx + n$ is divisible by both $(x - 1)$ and $(x + 1)$.
4. Use the factor theorem to determine if $(2x - 1)$ is a factor of $P(x) = 4x^3 - 4x^2 + x$.
5. Given $P(x) = x^3 - 7x + 6$, find all integer roots by testing factors of the constant term.
6. Divide $x^3 - 2x^2 + x - 5$ by $(x - 3)$ using long division and state the quotient and remainder.
7. The polynomial $f(x) = ax^3 + 3x^2 + b$ has a factor of $(x - 1)$ and $f(2) = 27$. Find a and b.

8. If $P(x) = (x-k)Q(x) + R$, explain why $P(k) = R$.

9. Factorise $P(x) = x^3 - 19x - 30$ completely, given that $P(-2) = 0$.

10. A polynomial $P(x)$ leaves a remainder of 5 when divided by $(x-1)$ and a remainder of -1 when divided by $(x+2)$. Find the remainder when $P(x)$ is divided by $(x-1)(x+2)$.

Easier Exam Questions

1. (Knox 2022 Q4) What is the remainder when $P(x) = -x^3 - 2x^2 - 3x + 8$ is divided by $x+2$?

   (A) -14

   (B) -2

   (C) 2

   (D) 14

2. (SCEGGS 2023 Q8c) The polynomial $P(x) = 2x^3 - x^2 - 16x + a$ has $(x+3)$ as a factor. **1**

   (i) Find the value of a.

   (ii) Hence, or otherwise, factorise $P(x)$ into linear factors. **1**

3. (Glenwood 2020 Q7e) Consider the polynomial $P(x) = 2x^3 + 3x^2 - 29x - 60$.

   i) Use the factor theorem to show that $(x+3)$ is a factor of $P(x)$ **1**

   ii) Factorise $P(x)$ completely. **2**

4. (Fort St 2022 Q7b) Let $P(x) = 3x^3 + 2x^2 - 7x + 2$.

   (i) Show that $(x-1)$ is a factor of $P(x)$. **1**

   (ii) Hence express $P(x)$ as a product of three linear factors. **2**

5. (Sydney Grammar 2019 Q9a) Find the remainder when the polynomial $P(x) = x^3 - 2x^2 + x + 5$ is divided by $(x-2)$. **1**

6. (SCEGGS 2019 Q9 a,b & c) Let $f(x) = x^2 + 5$ and $g(x) = -4x$.

   (a) Find $g(x) - f(x)$. **1**

   (b) What is the degree of the polynomial $f(x) \times g(x)$? **1**

   (c) Find $f(g(x))$. **1**

7. (Moriah 2021 Q7b) Consider the polynomial $P(x) = x^3 - 6x^2 + 10x - 3$.

   (i) Show that $P(3) = 0$ **1**

   (ii) Find all solutions to the polynomial equation $P(x) = 0$, leaving answers in exact form **3**

Harder Exam Questions

1. (Fort St 2021 Q1b) $P(x) = x^3 + 2x^2 - 5x + 1$.

   (i) Without using polynomial division, show that $P(x) \div (x-2)$ has remainder 7. **1**

   (ii) Verify the result of (i) using polynomial division. **3**

2. (Hurlstone 2024 Q15a & b) For what value of m is $2x^4 + mx^2 - 2$ divisible by $x + 2$? **1**

   (i) Use long division to divide the polynomial $P(x) = x^4 - x^3 + x^2 - x + 1$ by the polynomial $x^2 + 4$. **2**

   (ii) Hence, find the values of a and b such that $x^4 - x^3 + x^2 + ax + b$ is divisible by $x^2 + 4$. **2**

3. (Sydney Tech 2020 Q12b) A polynomial $f(x)$ is given by the equation $f(x) = x(x+1) - a(a+1)$ for some constant a.

   i. By using the factor theorem, find one factor of $f(x)$ in terms of a. **1**

   ii. By division, or otherwise, express $f(x)$ as a product of linear factors. **2**

4. (Presbyterian 2021 Q7a) Let $P(x) = x^3 - 3x^2 + kx - 5$, where k is a rational number. **2**

   When $P(x)$ is divided by $(x + 2)$, the remainder is 7. Find k.

5. (Gosford 2022 Q1a) The polynomial $P(x) = x^4 - 2x^3 - kx + 27$ has a factor of $(x - 2)$ as a factor. **2**

   What is the value of k?

6. (Baulkham Hills 2019 Q11c) (i) Find the linear factors of $x^3 - 5x^2 + 8x - 4$ **2**

   (ii) Hence solve $x^3 - 5x^2 + 8x - 4 > 0$ **2**

7. (Knox 2023 Q11b) The polynomial $P(x) = 2x^3 + ax + 4$ has a zero at $x = -2$. **2**

   Find the value of a.

8. (Knox 2023 Q11d) (i) Sketch the polynomial $y = x(x + 3)(x - 2)^2$. **2**

   (ii) Hence, solve the inequation $x(x + 3)(x - 2)^2 \geq 0$. **1**

9. (Sydney Tech 2019 Q13b) The polynomial $P(x) = x^2 + ax + b$ has a zero at $x = 2$ and when $P(x)$ is divided by $x + 1$, the remainder is 18. Find the values of a and b. **2**

10. (Baulkham Hills 2022 Q13b) Consider the polynomial $f(x) = px^3 - 4x^2 - 11x + q$ where p and q are constants. **4**

    Given that $(x - 2)$ and $(2x + 3)$ are both factors of $f(x)$, find the values of p and q, and hence fully factorise $f(x)$.

11. (Baulkham Hills 2023 Q14e) Let $P(x) = x^2 - 2ax + a^4$ where a is a non-zero real number. **3**

    When $P(x)$ is divided by $x + b$, the remainder is 1.

    The polynomial $Q(x) = bx^2 + x + 1$ has a factor of $(ax - 1)$.

    Determine the value(s) of b.

12. (North Sydney Boys 2023 Q2) Show that $(x - 1)(x - 2)$ is a factor of $P(x) = x^n(2^m - 1) + x^m(1 - 2^n) + (2^n - 2^m)$ where **2**

    m and n are positive integers.

13. (Caringbah 2019 Q14 a)i) The polynomial $P(x) = x^3 + ax^2 + bx + 6$ has $(x - 2)$ as a factor. When $P(x)$ is divided by $(x - 1)$, the remainder is 10. Find the values of a and b. **2**

14. (Glenwood 2024 Q7e) The polynomial $P(x) = x^3 + x^2 + ax + b$ has a factor of $(x - 1)$. When $P(x)$ is divided by $(x - 2)$, the remainder is 10. Find the values of a and b. **2**